# MILITARY SPECIFICATION

## LINK, CARTRIDGE, METALLIC BELT, 30MM, M29

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification for the M29 Link covers one type of metallic belt link intended to hold M788 and M789 30 millimeter ammunition in the form of flexible belts for use in the ADEN MK IV, 30MM and M230 gun systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-W-13855 | - | Weapons, Small Arms and Aircraft Armament |
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specification for |
| MIL-W-63150 | - | Standard Quality Assurance Provisions for Weapons and Support Material |

STANDARDS

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspections by Attributes. |

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| | | |
|---|---|---|
| MIL-STD-331 | - | Fuze and Fuze Components, Environmental and Performance Tests For |
| MIL-STD-644 | - | Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition |

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

| | | |
|---|---|---|
| 11825879 | - | Link, Cartridge, 30MM - M29 |
| 11825934 | - | Cartridge, 30MM, Dummy XM848 |
| 11825891 | - | Shipping and Storage, 30MM Linked, RH - Packed |
| 11825892 | - | Packing, Cartridge, 30MM Linked Left Hand, Shipping & Storage Container |
| 11825893 | - | Marking on Shipping and Storage Container, M592, 30MM Cartridge |

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adapted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable:

STANDARDS

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | | |
|---|---|---|
| ASTM E 18 | - | Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of the conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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## 3. REQUIREMENTS

3.1 Material. Material shall be in accordance with applicable drawings and specifications.

3.2 Link. The link shall comply with all requirements specified on Drawing (Dwg.) 11825879, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Manufacturing process. The links shall be manufactured by the identical process used for first article sample and no deviation from the process shall be made without prior approval of the contracting officer.

3.4 Hardness. The link shall comply with the hardness requirements specified on Dwg. 11825879.

3.5 Pull and elongation. A tensile load of 227 kg shall be applied gradually along the lateral axis of the link. After the load is removed, the increase in pitch shall not exceed 0.75 millimeters. Testing shall be as specified in 4.5.2.

3.6 Cartridge stripping. The force required to completely strip a dummy cartridge from a link using a centrally applied load (without impact) shall be not less than 31.75 Kgs. nor more than 74.84 Kgs. Links shall assemble and disassemble without scratching or removing metal (shaving) from the cartridge cases.

3.7 Free hinging. When assembled with cartridges to form a belt, the links shall freely pivot at the hook/loop joint without binding or unhooking.

3.8 Functioning. The link shall be capable of being assembled with M788 and M789 cartridges into belts and cycled through a belt-fed ADEN MK IV gun system without cracking, breaking, separating or causing a malfunction. The firing rate shall be at the system design rate.

3.9 First article inspection. This specification contains technical provisions for first inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. The requirements for workmanshp shall be in accordance with MIL-W-63150, applicable drawings, referenced specifications and the following:

3.10.1 Metal defects. The link shall be free of blisters, seams, laminations, deep scratches and porosity. Cracks, or tears are not permitted in the radius leading into each loop of the link.



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3.10.2 Burrs. All surfaces of the link shall be free of any rough edge which could cause definite removal of metal from the cartridge case, injure personnel during handling, or result in binding of the link.

3.10.3 Foreign matter. No link shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.



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TABLE I. First article inspection.

**CLASSIFICATION OF DEFECTS & TESTS** DOD-L-63983(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 11885879 |
|---|---|---|---|---|---|
| | Link, Cartridge, 30MM - M29 | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | | No. of Liner | | | |
| | Examination for defects | 150 | ACC-0 REJ-1 | 3.2 | 4.4.2.1 |
| | Hardness | 32 | ACC-0 REJ-1 | 3.4 | Test Fixture |
| | Corrosion resistance | 32 | ACC-0 REJ-1 | 3.2 | 4.5.5 |
| | Free hinging | 3 Belts | ACC-0 REJ-1 | 3.7 | 4.5.1 |
| | Stripping | 32 | ACC-1 REJ-2 | 3.6 | 4.5.3 |
| | Elongation | 32 | ACC-1 REJ-2 | 3.5 | 4.5.2 |
| | Functioning | 6 Belts | ACC-0 REJ-1 | 3.8 | 4.5.4 |

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall not exceed 500,000 links.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

DOD-L-63983(AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Link, Cartridge, 30MM - M29 | 1 OF 2 | 11885879 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL: | None defined | See 4.4.2a | | | |
| MAJOR: | | | | | |
| 101. | Width of eye, max. 1/ | | .65% | 3.2 | Gage |
| 102. | Length of eye, max. 1/ | | .65% | 3.2 | Gage |
| 103. | Contour of eye, min. | | .65% | 3.2 | Gage |
| 104. | Pitch, max. | | .65% | 3.2 | Gage |
| 105. | Functional diameter | | .65% | 3.2 | Gage |
| 106. | Phosphate coating | | .65% | 3.2 | Test Panels |
| 107. | Free hinging | | .65% | 3.2 | 4.5.1 |
| MINOR: | | | | | |
| 201. | Centerline to the shelf 1/ | | 1.0% | 3.2 | Gage |
| 202. | Centerline to outer surface, at eye 1/ | | 1.0% | 3.2 | Gage |
| 203. | Location, to eye 1/ | | 1.0% | 3.2 | Gage |
| 204. | Location, surface to eye 1/ | | 1.0% | 3.2 | Gage |
| 205. | True position of eye 1/ | | 1.0% | 3.2 | Gage |
| 206. | Distance, cartridge surface to loop opening 1/ | | 1.0% | 3.2 | Gage |
| 207. | Height, center shiffening | | 1.0% | 3.2 | Gage |
| 208. | Distance, centerline to hook end | | 1.0% | 3.2 | Gage |
| 209. | Distance, cartridge surface to hook end 1/ | | 1.0% | 3.2 | Gage |

NOTES:

1/ Quality conformance inspection shall be conducted on the noted characteristic when there is a change in production tooling.

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER 11885879 |
|---|---|---|---|---|---|
| 4.4.2.1 | Link, Cartridge, 30MM - M29 | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | | See 4.4.2a | | | |
| 210. | Overall length of link | | 1.0% | 3.2 | Gage |
| 211. | Height of stiffening rib on eye | | 1.0% | 3.2 | Gage |
| 212. | Missing or defective protective coating | | 1.5% | 3.2 | Visual |
| 213. | Improper or missing identification | | 2.5% | 3.2 | Visual |
| 214. | Workmanship | | 2.5% | 3.10 | Visual |
| | TESTS | | | | |
| | Hardness | 32 | ACC-0 REJ-1 | 3.4 | Test Fixture |
| | Corrosion resistance | 32 | ACC-0 REJ-1 | 3.2 | 4.5.5 |
| | Free hinging | 3 Belts | ACC-0 REJ-1 | 3.7 | 4.5.1 |
| | Elongation | 32 | ACC-1 REJ-1 | 3.5 | 4.5.2 |
| | Functioning | 6 Belts | ACC-0 REJ-1 | 3.8 | 4.5.4 |

NOTES:

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4.4.3 Testing.

4.4.3.1 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.6 herein.

4.5 Test methods and procedures.

4.5.1 Free hinging. The sample links shall be tested by assembling the links into belts of 72 links each using standard XM848 dummy cartridges as specified on Dwg. 11825934. Each belt shall be laid out horizontally to its full length, with the open side of the links up. One end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links down) position. One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position.

4.5.2 Pull and elongation. Each link shall be inserted in an approved test fixture and a 5 $\pm$ 1 kg lateral static load shall be applied. With this load applied, the link pitch shall be measured and recorded. Then, gradually the load shall be increased to 227 (KG) by lateral load. After applying this load for one (1) minute, the pull load shall be released and the 5 $\pm$ 1 kg static load reapplied. The pitch distance shall be remeasured and recorded. Upon completion of this test, the sample links shall be scraped.

4.5.3 Cartridge stripping. Each link shall be placed in an approved holding fixture and the dummy cartridge Dwg. 11825934 shall be inserted into the link. An axial load shall be gradually applied along the longitudinal axis of the dummy at the base end. Observation shall be made of the load required to separate the cartridge from the link.

4.5.4 Function test. Links shall be tested by assembling the links into belts using M788 30MM cartridges and firing the belts from an ADEN 30MM weapon system including magazine. Test shall be conducted with the belted ammo conditioned at 25°C & 71°C.

4.5.5 Corrosion resistance. The links shall be subjected to the salt fog test of MIL-STD-810, Method 509, for a period of 48 hours.



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5. PACKAGING

5.1 Preservation, packaging, and packing. Links shall be preserved, packaged, packed and marked in accordance with MIL-STD-644 for the level of protection specified in the contract (see 6.1).

5.2 Marking. Marking for shipment shall be in accordance with the procurement document and shall comply with the requirements of Dwg. 11825893.

6. NOTES

6.1 Intended use. The metallic belt cartridge links are intended for use in the ADEN MK IV, and M230 30MM Gun Systems.

6.2 Ordering data. See MIL-A-48078.

6.3 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Contract data requirements. The requirements for records of inspection and testing will be listed directly on DD Form 1423 Contract Data Requirements List, incorporated into the contract.

6.5 Definition of terms.

6.5.1 Deformed link. A link whose shape or dimensions, or both, have been mutilated or altered to the extent that the links are unsuitable for their intended use.

6.5.2 Cracks. See Figures 1, 2 and 3.

6.5.3 Tears. Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth with length equal to or greater than the material stock thickness.

6.6 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

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